

What is the hydrogen ion concentration in the following household solutions?

- (a) household ammonia: $\text{pH} = 11.00$
- (b) vinegar: $\text{pH} = 2.0$
- (c) soda pop: $\text{pH} = 4.00$
- (d) drain cleaner: $\text{pH} = 14.00$

Express the following typical concentrations as pH values.

- (a) grapefruit juice: $[\text{H}_{(\text{aq})}^+] = 1.0 \times 10^{-3} \text{ mol/L}$
- (b) rainwater: $[\text{H}_{(\text{aq})}^+] = 1 \times 10^{-5} \text{ mol/L}$
- (c) milk: $[\text{H}_{(\text{aq})}^+] = 1 \times 10^{-7} \text{ mol/L}$
- (d) liquid soap: $[\text{H}_{(\text{aq})}^+] = 1.0 \times 10^{-10} \text{ mol/L}$